

SUMMER INTERNSHIP PROJECT REPORT

on

ARTIFICIAL INTELLIGENCE

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology (B.Tech)

in

Computer Science and Engineering

Submitted To:

IILM University, Greater Noida

Submitted By:

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Section: 2CSE7

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DECLARATION

I, Prince Maurya, student of B.Tech Computer Science and Engineering (Section 2CSE7) at IILM University, Greater Noida, hereby declare that the internship report entitled '**Artificial Intelligence**' is an original work conducted by me during my internship at **CodSoft**. The project tasks, including the Rule-Based Chatbot, Tic-Tac-Toe AI, and Advanced Face Detection modules, were implemented and documented by me as part of the internship completion criteria.

Date: _____

Signature: _____

Prince Maurya (25SCS1003001538)

ACKNOWLEDGEMENT

I would like to express my profound gratitude to **IILM University, Greater Noida**, and the Department of Computer Science and Engineering for providing me with the academic foundation to undertake this program.

I extend my sincere thanks to the team at **CodSoft** for giving me the platform to bridge the gap between theoretical computer science concepts and practical artificial intelligence applications. The structured tasks provided during this virtual internship were instrumental in honing my programming, machine learning, and problem-solving skills.

EXECUTIVE SUMMARY

This report documents the practical development work completed during the Artificial Intelligence internship at CodSoft. To fulfill the internship requirements, I developed three functional AI applications, each focusing on a distinct subfield of computer science:

- **Task 1 (NLP):** A Rule-Based Chatbot built in Python utilizing regular expressions to classify user intents and simulate responsive text interactions.
- **Task 2 (Game Theory):** An unbeatable Tic-Tac-Toe AI engineered using the recursive Minimax algorithm to execute advanced state-space evaluation.
- **Task 5 (Computer Vision):** A deep learning pipeline implementing the MTCNN architecture via OpenCV to dynamically preprocess images and execute high-accuracy facial detection.

PROJECT IMPLEMENTATION DETAILS

1. Rule-Based Conversational Chatbot

The objective was to create a conversational agent capable of understanding standard user inputs and responding contextually. The system normalizes raw text strings and uses Python's *re* module for precise pattern matching. Instead of rigid exact-string checks, the implementation utilizes word boundaries to identify core intents within broader sentences.

2. Unbeatable Tic-Tac-Toe AI Engine

This project involved implementing game theory principles to create a mathematically flawless AI opponent. The core logic relies on the Minimax algorithm, which recursively evaluates all possible future board states. By treating the human player as a minimizing agent and the AI as a maximizing agent, the algorithm maps out decision trees down to terminal states (win, loss, or draw) to select the optimal move.

3. Deep Learning Face Detection

Utilizing the Multi-task Cascaded Convolutional Networks (MTCNN) framework in conjunction with OpenCV, a robust face detection pipeline was constructed. The code pre-processes input images by resizing large formats and applying Gaussian sharpening. The neural network evaluates the RGB matrices to detect facial boundaries and extract specific landmarks, applying a strict 90% confidence threshold to filter out false positives.

CONCLUSION & LEARNING OUTCOMES

The Artificial Intelligence internship provided a valuable transition from academic theory into practical software engineering. Through iterative development, debugging, and continuous integration, I acquired several core competencies:

- **Algorithm Optimization:** Deepened understanding of recursive decision-making and search space optimization via the Minimax algorithm.
- **Data Processing:** Gained proficiency in text normalization, regular expression parsing, and multidimensional image array manipulation.
- **Deep Learning Integration:** Successfully integrated a pre-trained neural network (MTCNN) into a broader pipeline to execute complex vision tasks.
- **Professional Documentation:** Maintained clean, organized version control by publishing well-documented source code across dedicated GitHub repositories.

These skills directly complement the B.Tech Computer Science and Engineering curriculum at IILM University, establishing a robust technical foundation for my future career in Data Science and Machine Learning.